IMMANUEL SOUNTHONEVICHIT

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SUMMARY

Mechanical Engineering with Robotics student (on track for a 1st) seeking an engineering placement year. Mechanically proficient (CAD, manufacturing, mechanics, thermofluids) with hands-on experience in prototyping, workshop manufacturing, electronics and embedded hardware. Experienced in programming in C and Python, and in using MATLAB and Excel to collect and analyse data. Quick to learn and keen to take ownership of work.

EXPERIENCE

- **Manufacturing Workshop** Nov 2025
  University of Sussex *Brighton, UK*
  - Machined a novelty hammer to drawings on lathes and mills, following workshop safety procedures.
- **CAD Modelling** Feb 2025
  University of Sussex *Brighton, UK*
  - Modelled a record player in SolidWorks and produced dimensioned technical drawings.
  - Collaborated in a team of 4, dividing tasks to deliver the model efficiently.
- **Private Tutoring** 2023
  Independent *London, UK*
  - Taught maths and English, adapting explanations to each student's level.

PROJECTS

- **Linux Systems** |*Gentoo, OpenRC, Guix* Ongoing
  - Configure and maintain Gentoo and Arch Linux on a desktop and a ThinkPad X230, including hardened kernels, Portage, sysctl tuning, and Firejail/AppArmor sandboxing.
  - Diagnose and fix driver and kernel configuration issues. Currently learning Guile to build a GNU Guix System install.
- **Self-Hosted Homelab** |*Alpine Linux, WireGuard* Ongoing
  - Built a homelab on a Raspberry Pi 4 with secure remote access.
  - Configured the router and port forwarding to expose only required services.
- **HackSussex Hackathon** |*Godot* Feb 2026
  - Delivered a working rhythm game in 24 hours in a team of 3, splitting tasks and time-boxing to meet the deadline.
  - Debugged collaboratively through pair programming.
- **Global Design Challenge** |*Concept design* Jan 2026
  - Worked in a team of 5 to design an app for circulating second-hand items, balancing sustainability, user needs and practical constraints.
  - Pitched the concept to a panel and answered questions on the design decisions.
- **CanSat – Main Engineer** Nov 2024 – Feb 2025
  |*Python, Telemetry*
  - Led integration of all sensors and the telemetry programming for a can-sized satellite, delivered on deadline alongside A Levels.
  - Logged air pressure, temperature and humidity data in Python and analysed it to assess sensor performance.

SKILLS

- **Engineering**
  - SolidWorks, technical drawing, lathe and milling
- **Embedded & Hardware**
  - Sensor integration, Raspberry Pi
- **Software & Systems**
  - C, Python, C#, Shell, Git, Linux + GNU (Gentoo, Arch, Alpine, Guix), Godot, Unity, WireGuard
- **Data & Analysis**
  - MATLAB, Excel

EDUCATION

- **University of Sussex** Brighton, UK
  BEng Mechanical Engineering with Robotics (IMechE accredited) *Sep 2025 – Present*
  - 76% stage average
  - **Relevant modules:** Engineering Mechanics, Principles and Applications of Strength, Materials & Manufacturing Processes, Technical Drawing & CAD, Thermodynamics, Fluid Mechanics, Engineering Maths, Programming for Engineers, Electrical Circuits & Devices, Electrical Machines & Power Electronics
- **Wren Academy** London, UK
  A Levels: Computer Science, Physics, Maths *Sep 2023 – Jul 2025*
  - Developed an A Level Computer Science project in C# using Unity.